

Stage 5 Step-by-Step Practical & Conceptual Recap: Full OpenAPI & Swagger Verification

1. Architectural Synchronization (Resolving Path & Routing Mismatches)

- **Action Taken:** You updated your openapi.json blueprint and server.js route handlers to ensure exact path alignment.
 - Synchronized the paths to /tasks for collection operations (GET, POST).
 - Synchronized the paths to /tasks/{id} for item-level operations (GET, PUT, DELETE).
- **Browser Cache Purge:** Executed a Hard Refresh (Ctrl + F5) in Chrome at <http://localhost:3000/docs/> to flush outdated UI specs from browser memory and force Swagger UI to re-fetch the updated openapi.json file.
- **Sysadmin Analogy:** This step is equivalent to verifying that an Nginx or Apache reverse proxy routing table perfectly mirrors the actual backend application socket paths before opening the interface to live traffic.

2. Hands-On Interactive Validation in Swagger UI

You executed end-to-end CRUD verification directly through the interactive Swagger web interface (<http://localhost:3000/docs/>):

Step 1: Resource Creation (POST /tasks)

1. **Action:** Expanded the **POST /tasks** panel in Swagger UI and clicked **"Try it out"**.
2. **Payload Entry:** Submitted the JSON body:

```
{  "title": "Automate Kenyan SME invoices"}
```
3. **Execution:** Clicked **Execute**.
4. **Observed Response:**
 - **HTTP Status Code:** 201 Created
 - **cURL Command Generated:** `curl -X 'POST' 'http://localhost:3000/tasks' -H 'accept: */*' -H 'Content-Type: application/json' -d '{"title": "Automate Kenyan SME invoices"}'`
 - **Response Body:** Returned dynamic primary key allocation:

```
{  "id": 4,  "title": "Automate Kenyan SME invoices",
```

```
"done": false
}
```

5. **Technical Significance:** Confirmed the middleware (`express.json()`) correctly parsed the inbound body, assigned dynamic primary key id: 4, and instantiated the task in RAM memory.

Step 2: Specific Record Lookup (GET /tasks/{id})

1. **Action:** Expanded the **GET /tasks/{id}** panel in Swagger UI and clicked "Try it out".
2. **Parameter Input:** Set the path parameter id to 4.
3. **Execution:** Clicked **Execute**.
4. **Observed Response:**
 - **Request URL:** `http://localhost:3000/tasks/4`
 - **HTTP Status Code:** 200 OK
 - **Response Body:**

```
{
  "id": 4,
  "title": "Automate Kenyan SME invoices",
  "done": false
}
```
5. **Technical Significance:** Confirmed path parameter parsing (`parseInt(req.params.id, 10)`) successfully queried the active RAM array and returned the exact matching record.

Step 3: State Mutation & Modification (PUT /tasks/{id})

1. **Action:** Expanded the **PUT /tasks/{id}** panel in Swagger UI and clicked "Try it out".
2. **Parameter & Payload Input:** Set id to 4 and provided the JSON body:

```
{
  "title": "Automate Kenyan SME invoices",
  "done": true
}
```
3. **Execution:** Clicked **Execute**.
4. **Observed Response:**
 - **HTTP Status Code:** 200 OK
 - **Response Text:** Task updated
5. **Technical Significance:** Validated in-memory state mutation. The server located task 4, updated the boolean done parameter from false to true, and saved the modified record.

Step 4: Resource Destruction / Purge (DELETE /tasks/{id})

1. **Action:** Expanded the **DELETE /tasks/{id}** panel in Swagger UI and clicked "Try it out".
2. **Parameter Input:** Set id to 4.

3. **Execution:** Clicked **Execute**.
4. **Observed Response:**
 - **Request URL:** http://localhost:3000/tasks/4
 - **HTTP Status Code:** 204 No Content
 - **Response Text:** Task deleted
5. **Technical Significance:** Confirmed array index locating (.findIndex()) and memory removal (.splice()). The server responded with standard HTTP 204, indicating a clean, silent deletion.

3. Key Technical Takeaways & Summary

Stage Action	Method & Path	Expected Status	Result Verified	Sysadmin / Production Parallel
Ingestion	POST /tasks	201 Created	Task id: 4 created	Submitting invoice entry form into database
Query	GET /tasks/4	200 OK	Record returned	Index lookup on primary key ID
Update	PUT /tasks/4	200 OK	done: true applied	Modifying job record status flag
Purge	DELETE /tasks/4	204 No Content	Record removed	Unregistering job entry from memory